

Quality assessment of RNA sequencing data after KRAS suppression

WEEK 2 PROGRESS REPORT

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Background and knowledge gap

KRAS supports growth in pancreatic ductal adenocarcinoma (PDAC). RNA sequencing can identify expression changes after treatment with KRAS small interfering RNA (siRNA). Before comparing treatments, we must assess read quality and mapping suitability.

Objective

Assess read quality and prepare the paired files for reference-genome alignment.

Methods

The input inventory contains 34 compressed FASTQ files from 17 paired-end runs. We reviewed 26 FastQC 0.12.1 reports and the existing MultiQC 1.35 summary. On 22 September, we submitted eight missing raw-read quality checks and a complete MultiQC summary. We also submitted HISAT2 2.2.3 pilots against human hg38 for Control M1 and KRAS P1, using the first 1,000,000 pairs with unstranded, FR and RF settings.

FastQC checks across 26 files

Check	Pass	Warn	Fail	Files
Per-base quality	26	0	0	26
Per-tile quality	26	0	0	26
Per-read quality	26	0	0	26
Base composition	0	0	26	26
GC distribution	26	0	0	26
N content	26	0	0	26
Read length	26	0	0	26
Duplication	0	0	26	26
Overrepresented sequences	26	0	0	26
Adapters	26	0	0	26

Results

All 26 files contained 150-base reads, with 38,181,985–47,840,900 reads per file and 50%–52% GC. R1 and R2 counts matched within all 13 runs. Both mates passed quality checks. MultiQC included 22 files and estimated 57.0%–65.2% duplication; this range applies only to those 22 files.

Interpretation

Base-composition bias and repeated sequences are common in RNA sequencing because library preparation and unequal transcript abundance affect these measures. The Fail labels alone do not prove poor sequencing or justify removing duplicate reads. The quality and adapter results support proceeding to mapping for the reviewed files, while checking library complexity after alignment. They do not establish differential expression.

Remaining analysis

The new Galaxy jobs are awaiting results. This interim report therefore includes no mapping percentages. Four missing files were submitted for import from ENA runs SRR24828471 and SRR24828472. Next, check all 34 FastQC results and the complete MultiQC output, confirm library orientation, then run full alignments. Pilot subsets are not random samples. Separate run labels do not establish independent biological replicates. No gene counts or differential-expression results are available.

References

1. Galaxy FastQC reports and MultiQC dataset 108, reviewed 19 September 2026. New job submissions checked 22 September 2026.
2. Babraham Bioinformatics. FastQC documentation: Per Base Sequence Content and Duplicate Sequences. bioinformatics.babraham.ac.uk/projects/fastqc/
3. Klomp JA et al. Defining the KRAS- and ERK-dependent transcriptome in KRAS-mutant cancers. Science. 2024. doi:10.1126/science.adk0775.

Author contributions

Yichuan Lin organized the data and coordinated the analysis and preparation of this report.

Review signature _____ Date _____